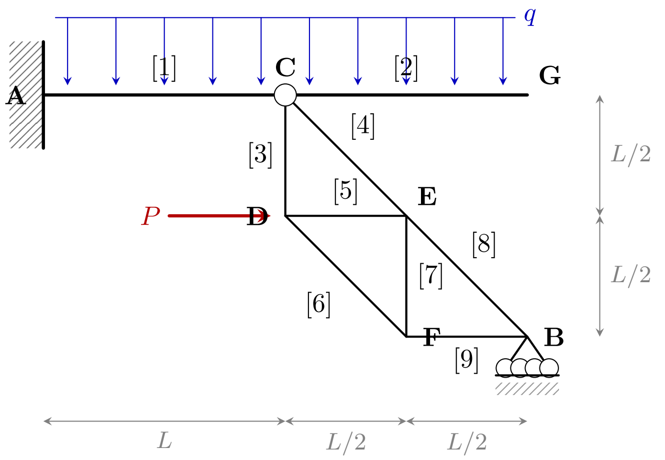
Model Solution

VU Mechanics – LAWI 100515 (PI) SS 2026

2nd Exercise Exam (26.06.2026) – Group A

System values: $L = 2\,\text{m}, \quad q = 5\,\text{kN/m}, \quad P = 10\,\text{kN}$

1. Task (70 %)



1.1 Static Determinacy

The structure consists of **two rigid bodies**: beam A–C–G (members [1],[2]) and the truss (nodes C,D,E,F,B; members [3]–[9]).

Overall system: 2 bodies $\Rightarrow$ $2 \times 3 = 6$ equilibrium conditions. Unknowns: support A (fixed, clamped) = 3, support B (roller) = 1, hinge C between beam and truss = 2. Total = 6.

$$n = \underbrace{3}_{A} + \underbrace{1}_{B} + \underbrace{2}_{C} - 2 \cdot 3 = 6 - 6 = 0 \quad \Rightarrow \quad \boxed{n = 0: \text{ statically determinate}}$$

1.2 Support Reactions (general form as functions of q, P, L)

Truss sub-system (joint forces C_x, C_y from beam on truss; B_V at B; P at D):

$$\sum M_C = 0: \quad P \cdot \tfrac{L}{2} + B_V \cdot L = 0 \qquad \Rightarrow \quad \boxed{B_V = -\tfrac{P}{2}}$$

$$\sum F_x = 0: \quad C_x + P = 0 \qquad \Rightarrow \quad C_x = -P$$

$$\sum F_y = 0: \quad C_y + B_V = 0 \qquad \Rightarrow \quad C_y = +\tfrac{P}{2}$$

Entire system (resultant distributed load $Q = 2qL$ at $x = L$):

$$\sum F_x = 0: \quad A_H + P = 0 \qquad \Rightarrow \quad \boxed{A_H = -P}$$

$$\sum F_y = 0: \quad A_V + B_V - 2qL = 0 \qquad \Rightarrow \quad \boxed{A_V = 2qL + \tfrac{P}{2}}$$

$$\sum M_A = 0: \quad M_A - 2qL \cdot L + P \cdot \tfrac{L}{2} + B_V \cdot 2L = 0 \qquad \Rightarrow \quad \boxed{M_A = 2qL^2 + \tfrac{PL}{2}}$$

1.3 Numerical Values

A_V	A_H	M_A	B_V
$2qL + \frac{P}{2} = \mathbf{25.00}\,\text{kN}$	$-P = \mathbf{-10.00}\,\text{kN}$	$2qL^2 + \frac{PL}{2} = \mathbf{50.00}\,\text{kNm}$	$-\frac{P}{2} = \mathbf{-5.00}\,\text{kN}$

Signs: positive ↑ for A_V, B_V; positive → for A_H; CCW for M_A. Negative ⇒ actual direction reversed.

1.4 Section Forces in Beams [1] and [2]

At node C the truss applies force $(-C_x, -C_y) = (+P, -\frac{P}{2})$ to the beam (rightward P, downward $\frac{P}{2}$). Running variable x measured from the left end of each beam.

Beam [1] (A → C), $0 \le x \le L$**:**

$$\begin{aligned} N_1(x) &= P \quad \text{(constant, tension)} \\ V_1(x) &= \left(2qL + \tfrac{P}{2}\right) - qx \\ M_1(x) &= \left(2qL + \tfrac{P}{2}\right)x - \tfrac{q}{2}x^2 - \left(2qL^2 + \tfrac{PL}{2}\right) \end{aligned}$$

Beam [2] (C → G), $0 \le x \le L$**:**

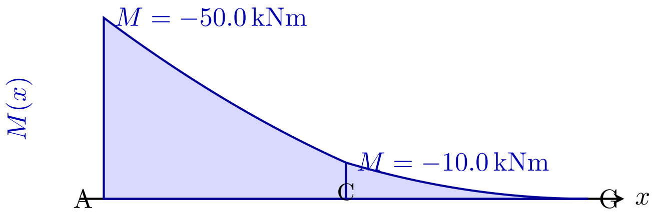
$$\begin{aligned} N_2(x) &= 0 \\ V_2(x) &= q(L - x) \\ M_2(x) &= -\tfrac{q}{2}(L - x)^2 \end{aligned}$$

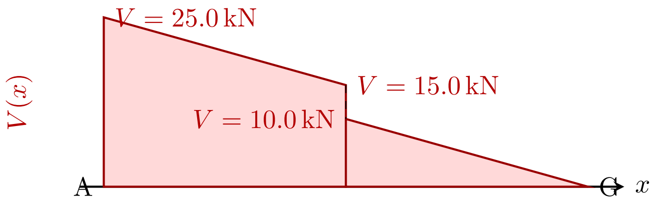
At C: shear jump $-\frac{P}{2}$ (vertical truss force); normal-force jump $-P$ (horizontal truss force); moment continuous but with kink.

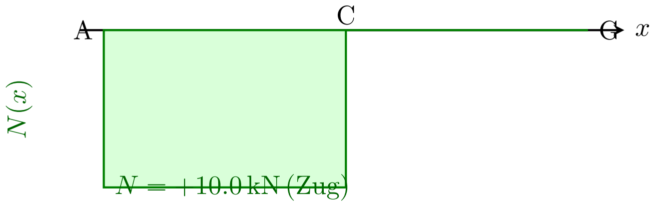
	Beam start			Beam end		
	M [kNm]	V [kN]	N [kN]	M [kNm]	V [kN]	N [kN]
Beam [1] (A→C)	−50.00	25.00	10.00	−10.00	15.00	10.00
Beam [2] (C→G)	−10.00	10.00	0.00	−0.00	0.00	0.00

1.5 Distributions $M(x)$, $V(x)$, $N(x)$

Schnittgrößenverläufe (auf der Zugseite; neg. M nach oben)







1.6 Axial Forces S_8 and S_9 – Circular Cut at Node B

At B: member [8] (B→E, direction $\frac{1}{\sqrt{2}}(-1,+1)$), member [9] (B→F, direction $(-1,\,0)$), support $B_V = -\frac{P}{2}$.

$$\sum F_y = 0: \quad \frac{S_8}{\sqrt{2}} + B_V = 0 \qquad \Rightarrow \quad \boxed{S_8 = \frac{P}{\sqrt{2}} \approx 7.07\,\text{kN (tension)}}$$

$$\sum F_x = 0: \quad -\frac{S_8}{\sqrt{2}} - S_9 = 0 \qquad \Rightarrow \quad \boxed{S_9 = -\frac{P}{2} = -5.00\,\text{kN (compression)}}$$

1.7 Axial Forces S_4, S_5, S_6 – Ritter's Cut

Section through [4],[5],[6]; right part {E,F,B} ($B_V = -\frac{P}{2}$ external; P in left part). Members [4] (C–E) and [6] (D–F): $-45°$; member [5] (D–E): horizontal.

S_5 – project all right-part forces onto $\frac{1}{\sqrt{2}}(1,1)$ (perpendicular to [4] and [6]):

$$-\frac{S_5}{\sqrt{2}} - \frac{P}{2\sqrt{2}} = 0 \;\Rightarrow\; \boxed{S_5 = -\frac{P}{2} = -5.00\,\text{kN (compression)}}$$

S_4 – moment sum about D ([5] and [6] pass through D):

$$\frac{S_4}{\sqrt{2}} \cdot \frac{L}{2} - \frac{P}{2} \cdot L = 0 \;\Rightarrow\; \boxed{S_4 = \sqrt{2}\,P = 14.14\,\text{kN (tension)}}$$

S_6 – moment sum about E ([4] and [5] pass through E):

$$-\frac{L}{2} \cdot \frac{S_6}{\sqrt{2}} - \frac{P}{2} \cdot \frac{L}{2} = 0 \;\Rightarrow\; \boxed{S_6 = -\frac{P}{\sqrt{2}} = -7.07\,\text{kN (compression)}}$$

S_8	S_9	S_4	S_5	S_6
$\frac{P}{\sqrt{2}} = 7.07$	$-\frac{P}{2} = -5.00$	$\sqrt{2}\,P = 14.14$	$-\frac{P}{2} = -5.00$	$-\frac{P}{\sqrt{2}} = -7.07$
[kN]; $+$ = tension, $-$ = compression				

2. Task (30 %)

Given: Simply supported beam (span L) with triangular load (0 at A, q_0 at B), $A_V = \frac{1}{6}q_0 L = 2.00\,\mathrm{kN}$, $B_V = \frac{1}{3}q_0 L = 4.00\,\mathrm{kN}$, and

$$M(x) = -\tfrac{q_0}{6L}\,x^3 + \tfrac{q_0 L}{6}\,x.$$

Values: $L = 2\,\mathrm{m}, \quad q_0 = 6\,\mathrm{kN/m}.$

2.1 Shear Force $V(x)$

$$V(x) = \frac{\mathrm{d}M}{\mathrm{d}x} = \frac{q_0 L}{6} - \frac{q_0}{2L}\,x^2$$

Check: $V(0) = \frac{1}{6}q_0 L = A_V$ ✓; $V(L) = -\frac{1}{3}q_0 L = -B_V$ ✓.

2.2 Location of Maximum Moment ($V = 0$)

$$\frac{q_0 L}{6} = \frac{q_0}{2L}\,x^2 \;\Rightarrow\; \boxed{x_{\max} = \frac{L}{\sqrt{3}} \approx 1.1547\,\mathrm{m}}$$

2.3 Maximum Moment

$$\boxed{M_{\max} = \frac{q_0 L^2}{9\sqrt{3}} = \frac{\sqrt{3}}{27}\,q_0 L^2 \approx 1.5396\,\mathrm{kNm}}$$

2.4 Distributions